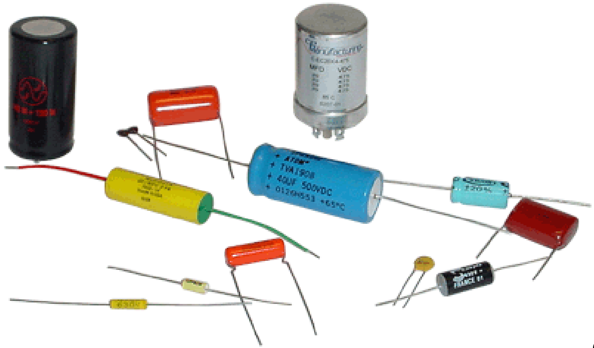


EBN 122

Chapter 6:
Capacitors and Inductors

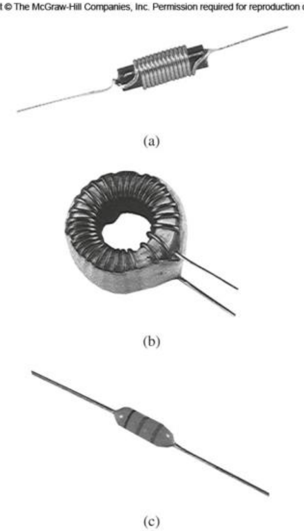
Inductors and Series and Parallel Inductors

Chapter 6 overview

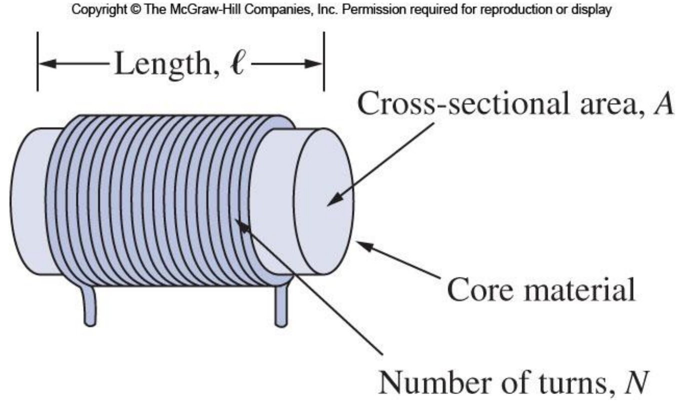
- ✓ 6.2. Capacitors
- ✓ 6.3. Series and Parallel Capacitors
- 6.4. Inductors
- 6.5. Series and Parallel Inductors

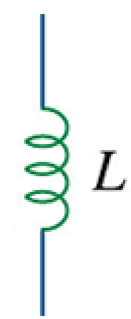
Inductors

- Passive element that stores energy in its **magnetic field**
- Any conductor of electric current has inductive properties
- Consists of coil with many turns of conducting wire around a core (iron, steel, air)



Courtesy of Tech America





Inductors

- If a current is allowed to pass through an inductor, the voltage v across the inductor is proportional to the time rate of change of current:

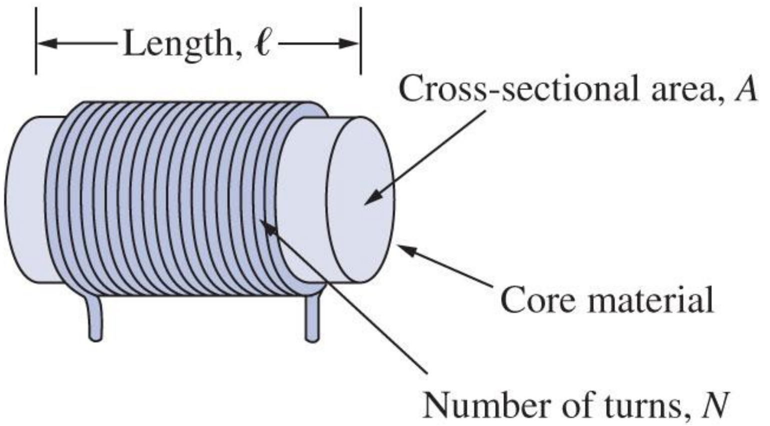
$$\boxed{v = L\frac{di}{dt}}$$

- **L = inductance** of inductor
- Inductance (L) is maeasured in henry (H)
- 1 henry = 1 volt-second per ampere

Inductors

- Inductance = opposition to the change of current flowing through an inductor
- L is dependent on **physical dimensions** of inductor, e.g. for solenoid:

$$\boxed{L = \frac{N^2 \mu A}{l} \;\; [H]}$$



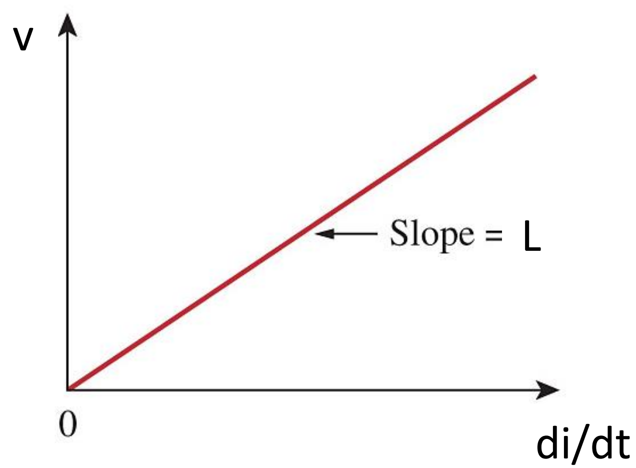
N = number of turns
A = cross sectional area of the core [m^2]
l = length of coil [m]
μ = permeability of the core [H/m]

Inductors

- Typical values = microhenrys (μH) to henrys (H)
- Fixed or variable

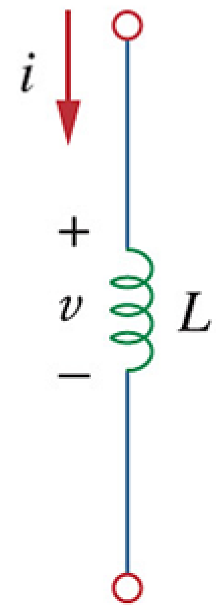
passive sign convention

- **Voltage-current** relationship of inductor:



$$\boxed{v = L\frac{di}{dt}}$$

Linear inductors
→ independent of current



Inductors

- **Current-voltage** relationship for inductor:

$$v = L\frac{di}{dt} \longrightarrow di = \frac{1}{L}v\,dt$$

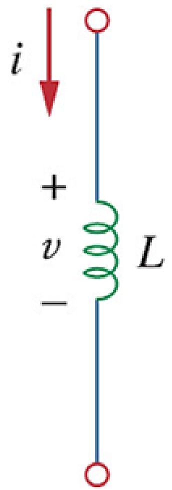
Integrating:

$$\boxed{i = \frac{1}{L}\int_{t_0}^{t} v(t)dt + i(t_0)}$$

$i(t_0)$ = total current for $-\infty < t < t_0$ and $i(-\infty) = 0$

→ $i(-\infty) = 0$ is reasonable as there was a time in the past when the inductor had no current

Inductors



- **Power (p)** delivered to inductor:

$$\boxed{p = vi = L\frac{di}{dt}i}$$

from $v = L\frac{di}{dt}$

- **Energy (w)** stored in inductor:

$$\boxed{w = \frac{1}{2}Li^2}$$

from $w = \int_{-\infty}^{t} p\,dt$

Inductors

Important properties of inductors:

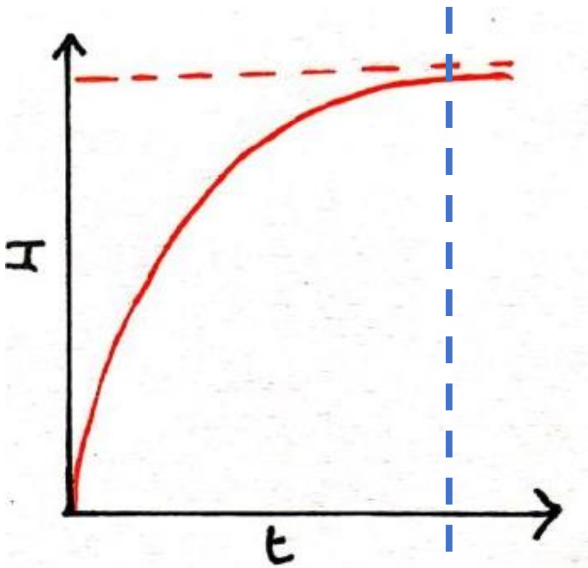
$$v = L\frac{di}{dt}$$

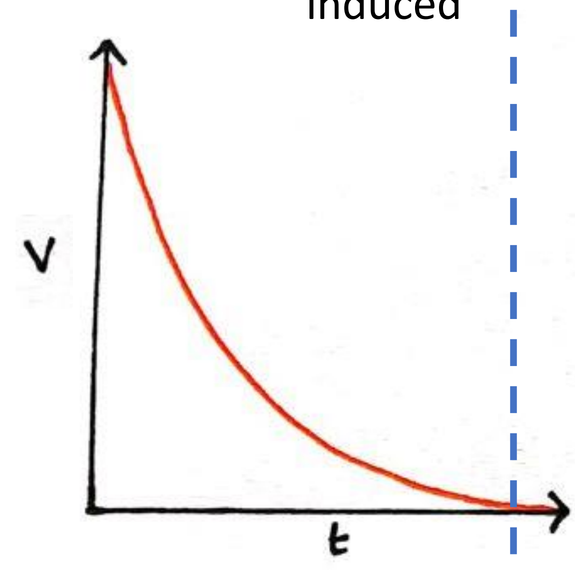
1. Inductor is a **short circuit** to dc
   (dc→ current constant → di/dt = 0
   → v = 0, thus short circuit)
2. **Current** through an inductor **cannot change instantaneously**
   (di/dt = ∞ → v = ∞, not possible)
3. Does not dissipate energy, but
   stores energy when **taking** power from circuit **(+p)**
   or
   returns previously stored energy when **delivering** power to circuit **(-p)**

Inductors

- Current and voltage over time:

2. When magnetic field at maximum, no more voltage is induced





1. Current flows through inductor and magnetic field builds

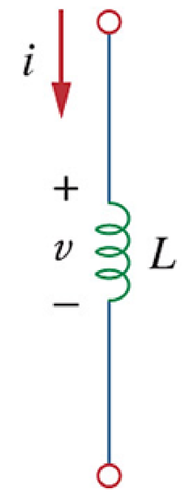
DC

DC

DC-state or steady-state

Inductors summary

$$L = \frac{N^2 \mu A}{l}$$



$$v = L\frac{di}{dt}$$

$$i = \frac{1}{L}\int_{t_0}^{t} v(t)dt + i(t_0)$$

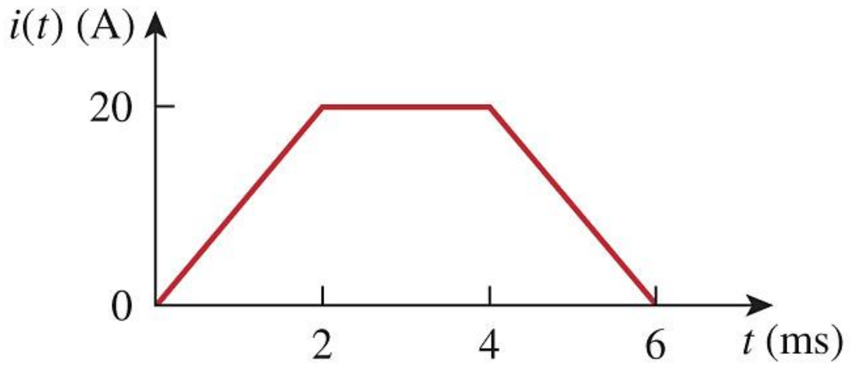
$$p = vi = L\frac{di}{dt}i$$

$$w = \frac{1}{2}Li^2$$

DC conditions:
short circuit

Example

The current through a 10 mH inductor is shown below. Determine the voltage across the inductor at t = 1, 3 and 5 ms.



Example

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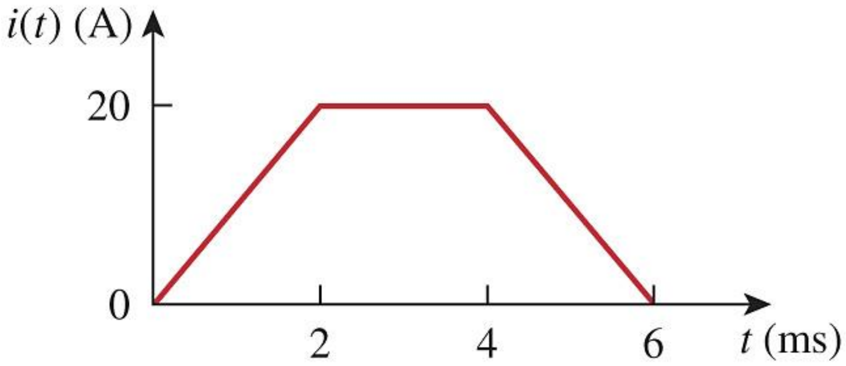


$$v(t) = 0.01\frac{di}{dt} = 0.01\frac{\Delta i}{\Delta t}$$

$$v(1ms) = 0.01\frac{20-0}{0.002\ \ -0} = \underline{100\text{ V}}$$

OR

$$i(t) = \frac{\Delta i}{\Delta t}t + c = \frac{20-0}{0.002\ \ -0}t + 0 = 10t\text{ kA}$$

$$v(t) = 0.01\frac{di}{dt} = 0.01 \times 10000 = 100\text{ V}$$

$$v(3ms) = \underline{0\text{ V}}\text{ because }\frac{di}{dt} = 0$$

$$v(5ms) = \underline{-100\text{ V}}$$

Practice Problem 6.9

Example 5: Inductor – voltage and current

The voltage across a 2 H inductor is $v(t) = 10(1 - t)$ V. Determine the current through the inductor at t = 4 s if i(0) = 2 A.

Practice Problem 6.9

Example 5: Inductor – voltage and current

The voltage across a 2 H inductor is $v(t) = 10(1 - t)$ V. Determine the current through the inductor at t = 4 s if i(0) = 2 A.

$$i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + \underbrace{i(t_0)}_{= 2\,A}$$

$$i(4) = \frac{1}{2}\int_0^4 \left[10(1-t)\right] + 2\,A$$

$$i(4) = 5\left(t - \frac{t^2}{2}\right)\Big|_0^4 + 2\,A$$

$$= 5t - \frac{5}{2}t^2\Big|_0^4 + 2A$$

$$= 5(4) - \frac{5}{2}(4^2) + 2A$$

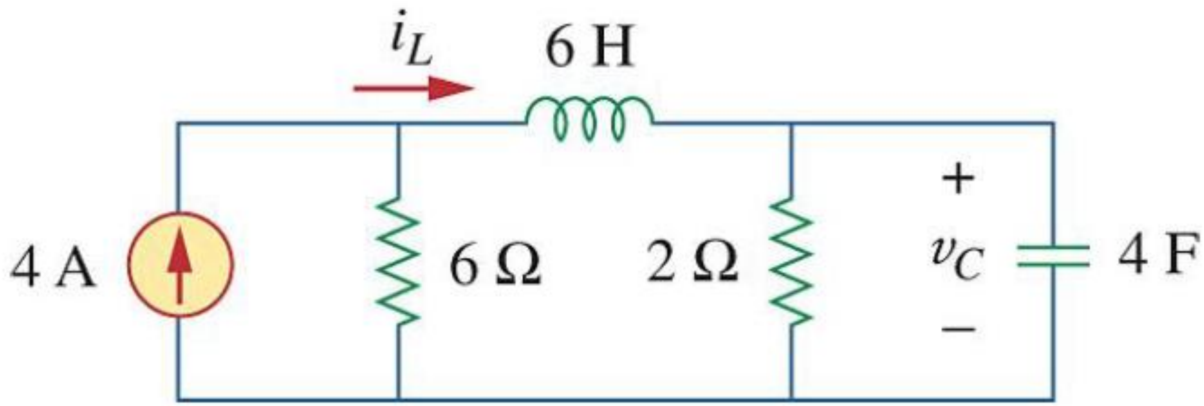
$$\therefore\ i(4) = 20 - 2.5(16) + 2$$

$$= 20 - 40 + 2$$

$$\therefore\ i(4) = \underline{-18\ A}$$

Example 6: Inductor – voltage, current and energy under DC condition

Determine v_C, i_L and the energy stored in both the capacitor and the inductor for the DC circuit below.



Example 6: Inductor – voltage, current and energy under DC condition

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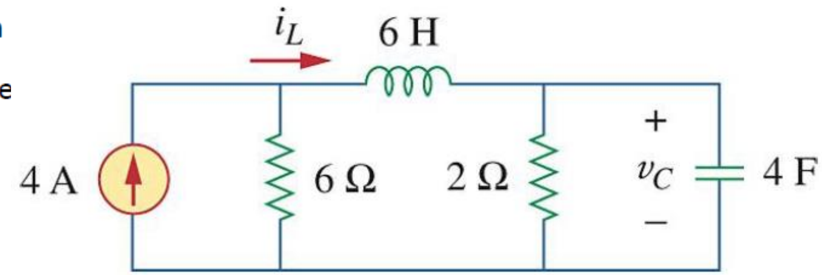
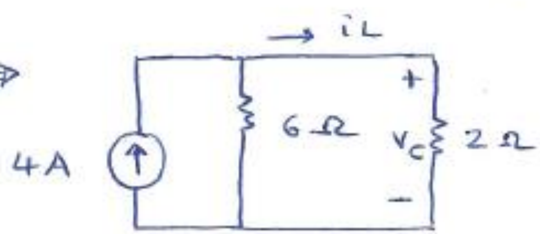


DC $\Rightarrow$ capacitor = open circuit, inductor = short circuit

$\Longrightarrow$



$$i_L = 4\left(\frac{6}{6+2}\right)$$ current division

$$\underline{i_L = 3\ A}$$

$$v_C = 2\,\Omega \times i_L$$

$$= 2 \times 3 = \underline{6\ V}$$

$$W_{capacitor} = \frac{1}{2} C v^2$$

$$= \frac{1}{2}(4F)(6^2) = \underline{72\ J}$$

$$W_{inductor} = \frac{1}{2} L i^2$$

$$= \frac{1}{2}(6H)(3^2) = \underline{27\ J}$$

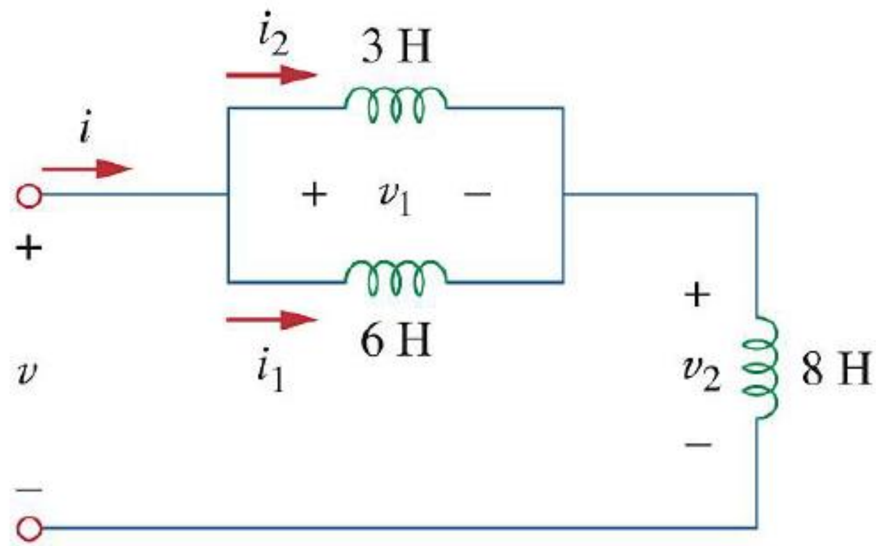
Example 7: Inductor – voltage and current

If $i_1(t) = 0.6e^{-2t}$ A and $i(0)$ = 1.4 A, find:

1) $i_2(0)$

2) $i_2(t)$ and $i(t)$, and

3) $v_1(t)$, $v_2(t)$ and $v(t)$



Example 7: Inductor – voltage and current

If $i_1(t) = 0.6e^{-2t}$ A and $i(0)$ = 1.4 A, find:

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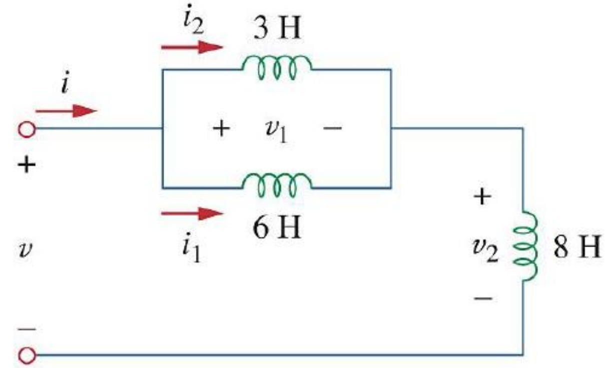


1) $i_1(0) = 0.6e^{-2(0)} = 0.6$ and KCL: $i(t) = i_1(t) + i_2(t)$

$\therefore i_2(0) = i(0) - i_1(0) \quad \therefore i_2(0) = 1.4 - 0.6 = 0.8$ A

2) $i_2(t) = \frac{1}{L}\int v_1\, dt + i_2(0)$ and $v_1(t) = L_1 \frac{di_1}{dt}$

$i_2(t) = \frac{1}{3}\int_0^t (-7.2e^{-2t}) + 0.8$ $\quad = 6\frac{d}{dt}(0.6e^{-2t})$

$= 6 \times (-2)(0.6e^{-2t})$

$\therefore i_2(t) = \frac{-7.2}{3}\left(-\frac{e^{-2t}}{2}\right)\Big|_0^t + 0.8$ $\quad v_1(t) = -7.2e^{-2t}$ V

$i_2(t) = 1.2(e^{-2t} - 1) + 0.8 = 1.2e^{-2t} - 0.4$ A

$i(t) = i_1(t) + i_2(t) = 0.6e^{-2t} + 1.2e^{-2t} - 0.4 = 1.8e^{-2t} - 0.4$ A

3) $v_2(t) = L\frac{di}{dt} = 8\frac{d}{dt}(1.8e^{-2t} - 0.4)$

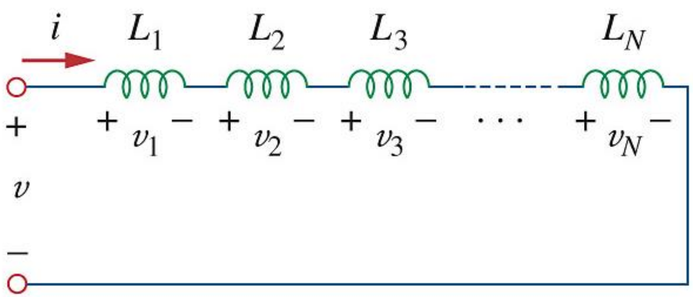
$= 8(-2(1.8)e^{-2t}) = -28.8e^{-2t}$

KVL: $v(t) = v_1(t) + v_2(t) = -7.2e^{-2t} + -28.8e^{-2t}$

$\therefore v(t) = -36e^{-2t}$ V

Series and parallel inductors

- Series and parallel combinations simplify circuits
- Equivalent inductance L_{eq} can be used



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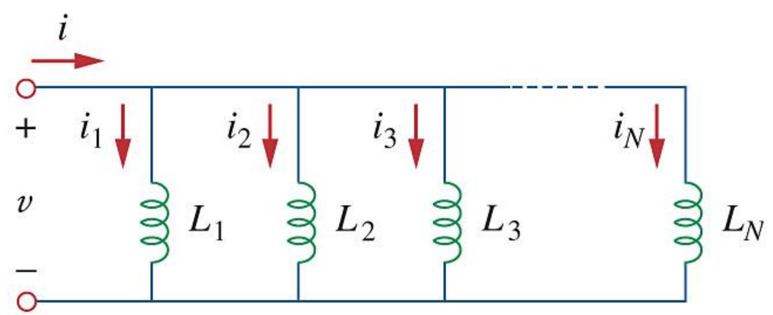
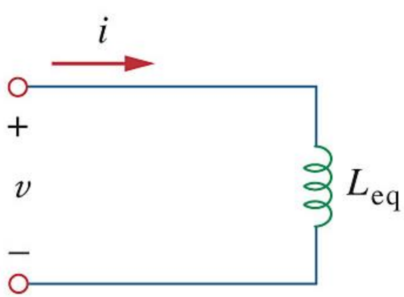
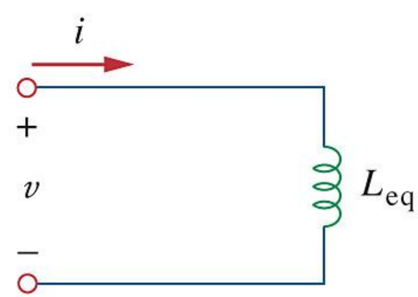


(a)



(b)



(b)

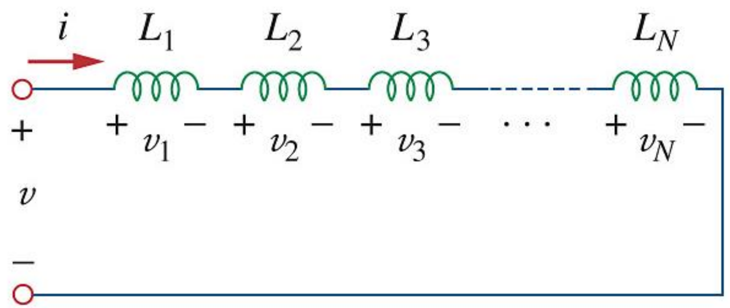
Series inductors

- Current through each inductor is the same
- Applying KVL:

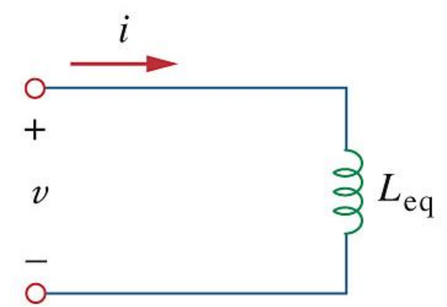
$$v = v_1 + v_2 + v_3 + \ldots + v_N$$

$$v = L_{eq}\frac{di}{dt} = L_1\frac{di}{dt} + L_2\frac{di}{dt} + L_3\frac{di}{dt} + \cdots + L_N\frac{di}{dt}$$

$$\boxed{L_{eq} = L_1 + L_2 + \ldots + L_N}$$



(a)



(b)

Series inductors
= sum of inductors

Parallel inductors

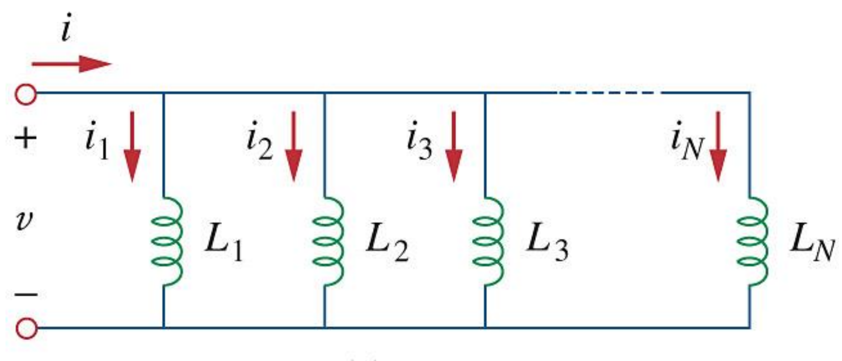
- Voltage across each inductor the same
- Applying KCL:

$$i = i_1 + i_2 + i_3 + \cdots + i_N$$

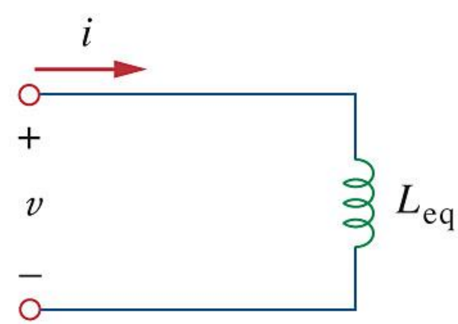
$$i = \frac{1}{L_{eq}} \int_{t_0}^{t} v dt + i(t_0) =$$

$$\frac{1}{L_1} \int_{t_0}^{t} v dt + i_1(t_0) + \ldots + \frac{1}{L_N} \int_{t_0}^{t} v dt + iN(t_0)$$

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}$$



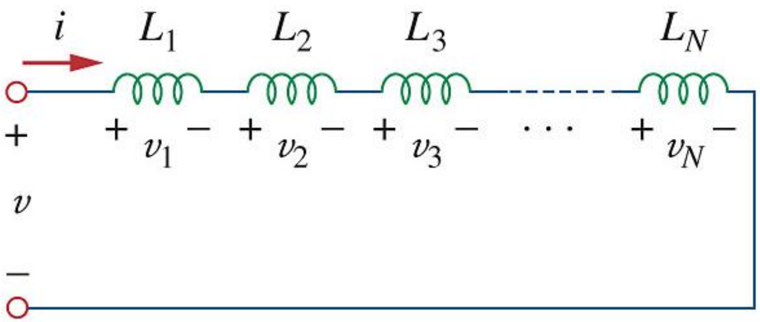
(a)



(b)

Parallel inductors
→ similar to parallel resistors

Series and parallel inductors



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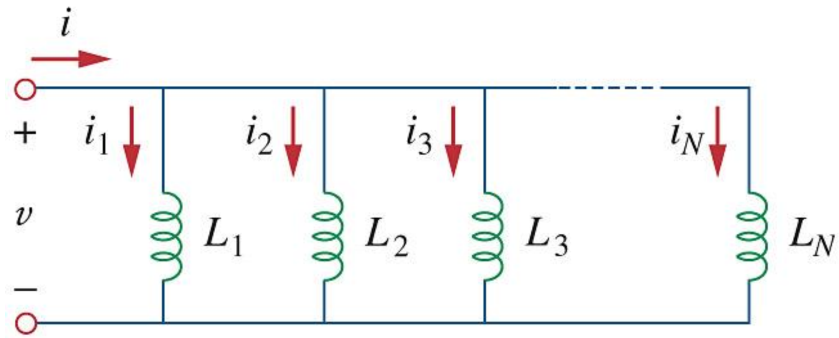
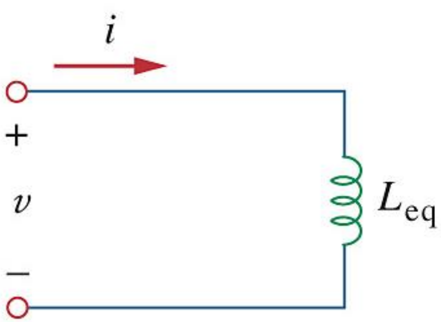


(a)



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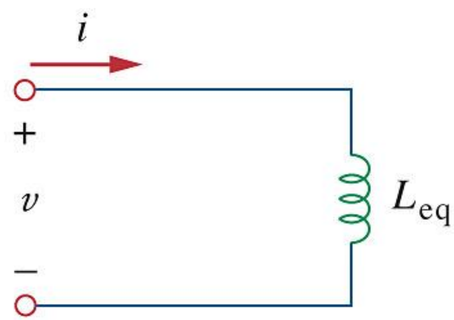


(b)

$$L_{eq} = L_1 + L_2 + \ldots + L_N$$

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}$$

Summary

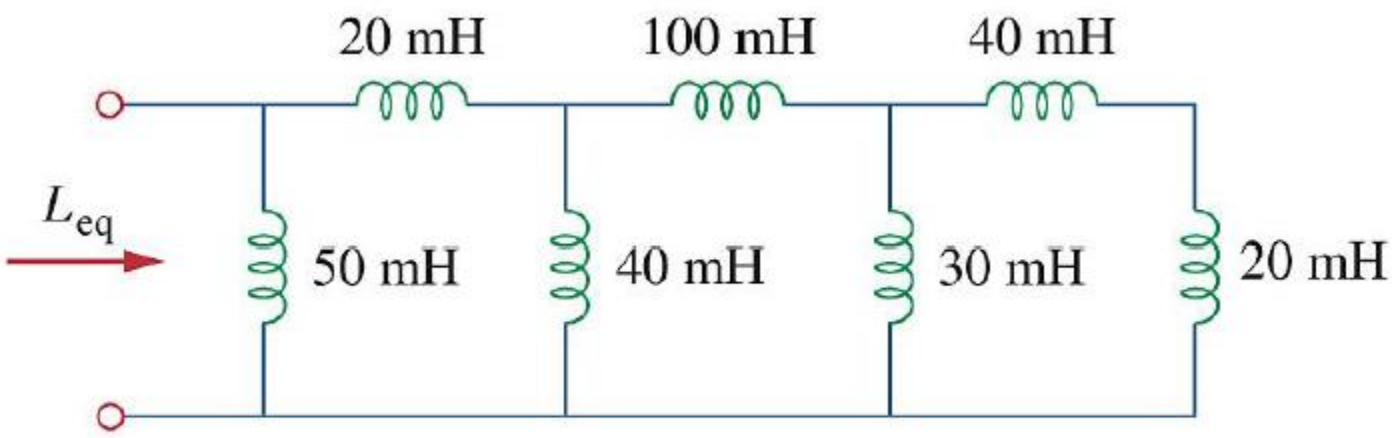
Important characteristics of the basic elements

Relation	Resistor (R)	Capacitor *(C)*	Inductor *(L)*
$v - i$:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
$i - v$:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
P or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	*v*	*i*

Example 8: Inductor – equivalent inductance

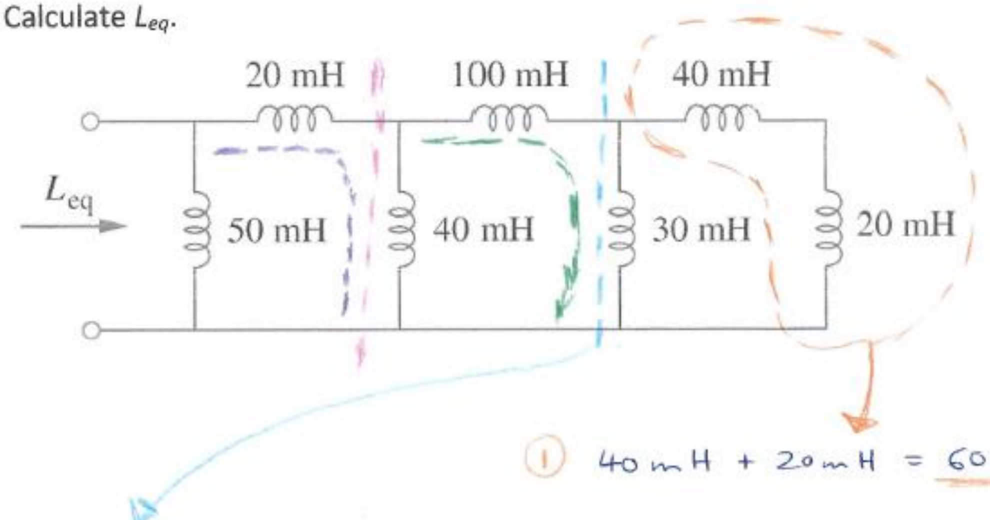
Practice Problem 6.11

Calculate L_{eq}.



Example 8: Inductor – equivalent inductance

Calculate L_{eq}.



① $40\,mH + 20\,mH = \underline{60\,mH}$

② $30\,mH \parallel \underline{60\,mH} = \dfrac{30 \times 60}{30 + 60} = \underline{20\,mH}$

③ $100\,mH + \underline{20\,mH} = \underline{120\,mH}$

④ $40\,mH \parallel \underline{120\,mH} = \dfrac{40 \times 120}{40 + 120} = \underline{30\,mH}$

⑤ $20\,mH + \underline{30\,mH} = \underline{50\,mH}$

$$L_{eq} = 50\,mH \parallel \underline{50\,mH}$$

$$= \frac{50 \times 50}{50 + 50} \qquad \therefore \; L_{eq} = \underline{25\,mH}$$